Understanding Copper Extraction

Where Copper is Found

Copper is usually found mixed with other elements in rocks, but sometimes its found pure. In Pakistan, copper is mined in places like North Waziristan, Chitral, Dir, Gilgit, Hazara, and parts of Quetta like Chaghi, Loralai, Pishin, and Zhob. Copper often appears in veins, grains, or scales within rocks.

The main copper ores are:

- **Sulphide Ores**: Like copper pyrite (CuFeS2) and copper glance (Cu2S).
- **Oxide Ores**: Like cuprite (Cu2O) and malachite (Cu2(OH)2CO3).

How Copper is Extracted from Sulphide Ores

Most copper (about 75%) comes from copper pyrite (CuFeS2) through a process called smelting. If the ore has at least 4% copper, smelting is used. For very low-quality ores, other methods are used. Heres how smelting works in simple steps:

Step 1: Concentration

The ore is crushed into a fine powder and mixed with water and a bit of pine oil. Air is blown through the mixture, making the copper sulphide particles stick to the oil and float to the top as froth. This froth is collected, leaving behind unwanted rocky material (gangue) at the bottom.

Step 2: Roasting

The concentrated ore is heated in a furnace with air. This burns off sulphur as sulphur dioxide gas (SO2) and removes impurities like arsenic and antimony as gases. Iron in the ore turns into iron oxide, and the ore now contains copper sulphide (Cu2S) and some iron sulphide.

Step 3: Smelting

The roasted ore is heated in a furnace with sand. The iron oxide combines with sand to form a waste product called slag (FeSiO3), which is removed. The copper sulphide is partially turned into copper oxide (Cu2O), which reacts with remaining copper sulphide to produce pure copper metal and more sulphur dioxide:

$$2Cu2S + 3O2- > 2Cu2O + 2SO2$$

$$2Cu2O + Cu2S- > 6Cu + SO2$$

Step 4: Bessemerization

The molten material is moved to a Bessemer converter, where air is blown through it. This oxidizes any remaining iron sulphide into iron oxide, which is removed as slag with added sand. The copper sulphide turns into copper oxide, which reacts to form metallic copper. Some copper oxide (CuO) is also formed, but its reduced back to copper using green wood poles, which release gases to help the process. The copper produced is called "blister copper" because it has a rough, bubbly surface. Its about 98% pure, with small amounts of iron, silver, or gold.

Step 5: Refining

Blister copper is purified further using an electrolytic process to make it almost 100% pure, but thats covered in another section.

Summary

Copper is mined from sulphide and oxide ores in places like Pakistan. To get copper from sulphide ores like copper pyrite, the ore is concentrated using froth flotation, roasted to remove impurities, smelted to separate copper from waste, and processed in a Bessemer converter to produce blister copper. This copper is then refined to make it pure for use.